

QM 2 -Continuous assessment II

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Question 1

0.0.1 (a)

We have our total Hamiltonian to be

$$\hat{H} = \hat{H}^0 + g\hat{V} = \begin{bmatrix} E_0 - g & 0 & 0 \\ 0 & 0 & g \\ 0 & g & E_0 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} E_0 - g - \lambda & 0 & 0 \\ 0 & 0 - \lambda & g \\ 0 & g & E_0 - \lambda \end{bmatrix} \quad (2)$$

$$[E_0 - g - \lambda] [(-\lambda)(E_0 - \lambda) - g^2] = 0 \quad (3)$$

$$\implies \lambda_0 = E_0 - g \quad (4)$$

$$\lambda^2 - E_0\lambda - g^2 = 0 \quad (5)$$

So we have $\lambda_0 = 0$ and a quadratic which we can solve.

$$\lambda_{\pm} = \frac{1}{2} \left[(E_0) \pm \sqrt{(E_0)^2 - 4(-g^2)} \right] \quad (6)$$

But we know the eigenvalues a time independent Hamiltonian are the energies

$$\epsilon_0 = E_0 - g \quad (7)$$

$$\epsilon_+ = \frac{1}{2} \left[(E_0) + \sqrt{(E_0)^2 + 4(g^2)} \right] \quad (8)$$

$$\epsilon_- = \frac{1}{2} \left[(E_0) - \sqrt{(E_0)^2 + 4(g^2)} \right] \quad (9)$$

Now we have the energy eigenvalues we can taylor expand for small g .

The $\epsilon_0 = g - E_0$ remains the same. For the taylor expansion we can use $(1+x)^\alpha = 1 + \alpha x$. So we can take out E_0

$$\pm E_0 \sqrt{1 + \frac{4g^2}{E_0^2}} \quad (10)$$

$$\approx \pm E_0 \left(1 + \frac{2g^2}{E_0^2} \right) + \dots \quad (11)$$

Thus we have

$$\epsilon_0 = E_0 - g \quad (12)$$

$$\epsilon_+ = \frac{1}{2} \left[E_0 + E_0 \left(1 + \frac{2g^2}{E_0^2} \right) \right] = E_0 + \frac{g^2}{E_0} + \dots \quad (13)$$

$$\epsilon_- = \frac{1}{2} \left[E_0 - E_0 \left(1 + \frac{2g^2}{E_0^2} \right) \right] = -\frac{g^2}{E_0} + \dots \quad (14)$$

Where if we need higher order terms we can use the general taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)^n \quad (15)$$

(b)

We see that we have two-fold degeneracy for $|1\rangle$ and $|3\rangle$ as they share the energy E_0 . $|2\rangle$ is unique (assuming $E_0 \neq 0$). So $|1\rangle$ and $|3\rangle$ give the degenerate subspace

$$\hat{V}_{deg} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \quad (16)$$

Which gives the characteristic equation

$$(-1 - \lambda)(-\lambda) = \lambda^2 + \lambda = 0 \quad (17)$$

$$\lambda = 0 \text{ and } \lambda = -1 \quad (18)$$

So we get the following up to first order for the degenerate states

$$\epsilon_0 = E_0 + (-1 \text{ or } 0)g \quad (19)$$

$$\epsilon_+ = E_0 \quad (20)$$

$$\epsilon_0 = E_0 - g \quad (21)$$

Non deg state

First order correction is given by

$$\epsilon_-^1 = \langle 2 | \hat{V} | 2 \rangle = 0 \quad (22)$$

This is evident if we go back into the vector representation of the states and multiply the row * matrix * column.

The second order correction in general is given by

$$\epsilon_-^2 = \sum_{m \neq n} \frac{|\langle m | H' | n \rangle|^2}{E_n^0 - E_m^0} = \frac{|\langle 1 | H' | 2 \rangle|^2}{E_2^0 - E_1^0} + \frac{|\langle 3 | H' | 2 \rangle|^2}{E_2^0 - E_3^0} \quad (23)$$

$$(24)$$

$$\epsilon_-^2 = \frac{|\langle 1 | -(|1\rangle\langle 1| + |2\rangle\langle 3| + |3\rangle\langle 2|) | 2 \rangle|^2}{E_2^0 - E_1^0} + \frac{|\langle 3 | (-|1\rangle\langle 1| + |2\rangle\langle 3| + |3\rangle\langle 2|) | 2 \rangle|^2}{E_2^0 - E_3^0} \quad (25)$$

$$(26)$$

$$\epsilon_-^2 = \frac{1}{-E_0} g^2 [|0|^2 + |1|^2] \quad (27)$$

$$\epsilon_-^2 = -\frac{g^2}{E_0} \quad (28)$$

Thus we get

$$\epsilon_-^2 = 0 + 0 + -\frac{g^2}{E_0} + \dots \quad (29)$$

Question 2

0.0.2 (a)

We have the 2D normalized wave function given by

$$\psi_{n_1, n_2}(x_1, x_2) = \frac{2}{a} \sin\left(\frac{n_1 \pi x_1}{a}\right) \sin\left(\frac{n_2 \pi x_2}{a}\right) \quad (30)$$

$$\text{with } n_1, n_2 \in \{1, 2, 3, \dots\} \quad (31)$$

Note here that as sine is periodic we see that for $\forall n_i$ even we get repetition the same result and likewise we get repetition of a different result $\forall n_i$ odd. Thus we can make life easier by considering the equivalent system with $n_1, n_2 \in \{1, 2\}$.

One could easily generalise from the 1D case which we have seen in class to find the Energy but I will derive it in case it is required.

We have the TISE

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi = E\psi \quad (32)$$

We notice that $\psi(x_1, x_2) = \psi(x_2, x_1)$ So we can take the derivative wrt only one coordinate then abuse this symmetry.

$$\frac{d\psi}{dx_1} = \frac{2}{a} \cos\left(\frac{n_1\pi x_1}{a}\right) \sin\left(\frac{n_2\pi x_2}{a}\right) \left(\frac{n_1\pi}{a}\right) \quad (33)$$

$$\frac{d^2\psi}{dx_1^2} = -\frac{2}{a} \sin\left(\frac{n_1\pi x_1}{a}\right) \sin\left(\frac{n_2\pi x_2}{a}\right) \left(\frac{n_1\pi}{a}\right)^2 \quad (34)$$

$$\implies \frac{d^2\psi}{dx_1^2} = -\left(\frac{n_1\pi}{a}\right)^2 \psi \quad (35)$$

Into Schrodinger's

$$\frac{\hbar^2}{2m} \left(\frac{n_1\pi}{a}\right)^2 \psi_{x_1} + V\psi_{x_1} = E_{x_1} \psi_{x_1} \quad (36)$$

Inside the well we have no potential so $V = 0$. Thus we have

$$\frac{\hbar^2}{2m} \left(\frac{n_1\pi}{a}\right)^2 = E_{x_1} \quad (37)$$

$$E = E_{x_1} + E_{x_2} \quad (38)$$

$$\implies E = \frac{\hbar^2}{2m} \left(\frac{n_1\pi}{a}\right)^2 + \frac{\hbar^2}{2m} \left(\frac{n_2\pi}{a}\right)^2 \quad (39)$$

We arrive at

$$E_{n_1, n_2} = \frac{\hbar^2 \pi^2}{2ma^2} (n_1^2 + n_2^2) \quad (40)$$

We can call $\alpha = \frac{\hbar^2 \pi^2}{2ma^2}$ We have to consider the cases **(a)** $(n_1, n_2) = (1, 1)$, **(b)** both cases $(n_1, n_2) = (1, 2)$ and $(n_1, n_2) = (2, 1)$, and **(c)** finally $(n_1, n_2) = (2, 2)$.

(a)

$$E_{(1,1)} = \frac{\hbar^2 \pi^2}{ma^2} \quad (41)$$

(b)

$$E_{(1,2)} = E_{(2,1)} = \alpha (1^2 + 2^2) = \alpha (2^2 + 1^2) = \frac{5\hbar^2 \pi^2}{2ma^2} \quad (42)$$

(c)

$$E_{(2,2)} = \alpha (2^2 + 2^2) = \frac{4\hbar^2 \pi^2}{ma^2} \quad (43)$$

We can see that $E_{(1,1)}$ and $E_{(2,2)}$ have no degeneracy while $E_{(1,2)}$ and $E_{(2,1)}$ have a two fold degeneracy.

(b)

We have the expression for the first order correction

$$E_{n_1, n_2}^1 = g \langle n_1, n_2 | V | n_1, n_2 \rangle \quad (44)$$

$$= g \int_0^a \int_0^a \psi_{n_1, n_2}^2(x_1, x_2) \left(x_1 - \frac{a}{2}\right) \left(x_2 - \frac{a}{2}\right) dx_1 dx_2 = g I_{n_1} I_{n_2} \quad (45)$$

We solve a general integral then sub in the indices

$$I_{n_i} = \frac{a}{2} \int_0^a \sin^2\left(\frac{n_i \pi x}{a}\right) \left(x - \frac{a}{2}\right) dx \quad (46)$$

$$\text{Let } u = x - \frac{a}{2} \quad (47)$$

$$I_{n_i} = \frac{a}{2} \int_{-\frac{a}{2}}^{\frac{a}{2}} \sin^2\left(n_i \pi \left(\frac{u}{a} + \frac{1}{2}\right)\right) u du \quad (48)$$

We see that $\sin^2(x)$ is even while x is odd. Thus our integrand is odd. We have the integral with symmetric bounds of an odd integrand which vanishes.

$$I_{n_i} = 0 \implies g I_{n_i} I_{n_i} = 0 \quad (49)$$

$$\implies E_{n_i, n_i}^1 = 0 \quad (50)$$

For the other case

$$E_{n_1, n_2} = \frac{4}{a^2} \int_0^a dx_1 \sin\left(\frac{2\pi x_1}{a}\right) \sin\left(\frac{\pi x_1}{a}\right) \left(x_1 - \frac{a}{2}\right) \int_0^a dx_2 \sin\left(\frac{2\pi x_2}{a}\right) \sin\left(\frac{\pi x_2}{a}\right) \left(x_2 - \frac{a}{2}\right) \quad (51)$$

$$I_i = \int_0^a dx_i \sin\left(\frac{2\pi x_i}{a}\right) \sin\left(\frac{\pi x_i}{a}\right) \left(x_i - \frac{a}{2}\right) \quad (52)$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)] \quad (53)$$

$$I_i = \frac{1}{2} \int_0^a \cos\left(\frac{\pi x_i}{a}\right) \left(x_i - \frac{a}{2}\right) dx_i - \frac{1}{2} \int_0^a \cos\left(\frac{3\pi x_i}{a}\right) \left(x_i - \frac{a}{2}\right) dx_i \quad (54)$$

$$u = x_i - \frac{a}{2}, \quad du = dx_i, \quad dv = \cos\left(\frac{n\pi x}{a}\right) dx_i \implies v = \frac{a}{n\pi} \sin\left(\frac{n\pi x}{a}\right) \quad (55)$$

$$I_n = \left[\left(x - \frac{a}{2}\right) \frac{a}{n\pi} \sin\left(\frac{n\pi x}{a}\right) \right]_0^a - \int_0^a \frac{a}{n\pi} \sin\left(\frac{n\pi x}{a}\right) dx \quad (56)$$

$$\text{first term vanishes at 0 and a } \int_0^a \sin\left(\frac{n\pi x}{a}\right) dx = \frac{a}{n\pi} \left[-\cos\left(\frac{n\pi x}{a}\right) \right]_0^a \quad (57)$$

$$= \frac{a}{n\pi} (1 - (-1)^n) \quad (58)$$

$$I_n = -\frac{a^2}{n^2 \pi^2} (1 - (-1)^n) \implies I_1 = \frac{-2a^2}{\pi^2}, \quad I_3 = -\frac{2a^2}{9\pi^2} \quad (59)$$

$$I_i = \frac{1}{2} I_1 + \frac{1}{2} I_3 = -\frac{8a^2}{9\pi^2} \text{ as we have no dependence on } i \text{ they are the same} \quad (60)$$

We have

$$\langle \psi_1 | V | \psi_3 \rangle = \langle \psi_3 | V | \psi_3 \rangle = \frac{4}{a^2} \left(-\frac{8a^2}{9\pi^2} \right)^2 = \frac{256a^2}{81\pi^4} \quad (61)$$

If we write the matrix

$$V = \begin{bmatrix} \langle \psi_1 | V | \psi_1 \rangle & \langle \psi_1 | V | \psi_3 \rangle \\ \langle \psi_3 | V | \psi_1 \rangle & \langle \psi_3 | V | \psi_3 \rangle \end{bmatrix} = \begin{bmatrix} 0 & \frac{256a^2}{81\pi^4} \\ \frac{256a^2}{81\pi^4} & 0 \end{bmatrix} \quad (62)$$

We have the first order correction

$$E_{1,3} = \frac{V_{11} + V_{22}}{2} \pm \frac{\sqrt{(V_{11} - V_{22})^2 + 4V_{12}V_{21}}}{2} \quad (63)$$

$$gE_{(1,2),(2,1)} = \pm g \frac{256a^2}{81\pi^4} \quad (64)$$

Question 3

0.0.3 (a)

We know that the variational energy is given by

$$E_{var} \leq \langle \psi_{var} | H | \psi_{var} \rangle = \langle H \rangle \quad (65)$$

As we have no time dependence we can identify the Hamiltonian operator

$$\hat{H}\psi = E\psi \quad (66)$$

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{m\omega^2 x^2}{2} + \lambda_4 x^4 \right) \psi = E\psi \quad (67)$$

$$\hat{T} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \text{ is the Kinetic operator.} \quad (68)$$

$$\hat{V} = \frac{m\omega^2 x^2}{2} + \lambda_4 x^4 \text{ is the Potential operator.} \quad (69)$$

We can identify the normalization

$$\int_{-\infty}^{\infty} |\psi_{var}|^2 dx = 1 \quad (70)$$

$$\int_{-\infty}^{\infty} \gamma^2(\alpha) \left(e^{-\alpha \frac{x^2}{2}} \right)^2 dx = 1 \quad (71)$$

Normalisation is independent of x . We have a standard Gaussian integral.

$$\gamma^2(\alpha) \sqrt{\frac{\pi}{\alpha}} = 1 \quad (72)$$

$$\implies \gamma = \left(\frac{\alpha}{\pi} \right)^{\frac{1}{4}} \quad (73)$$

We know $\langle \hat{H} \rangle = \langle \hat{T} \rangle + \langle \hat{V} \rangle$

$$\hat{T}\psi = \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \left[\left(\frac{\alpha}{\pi} \right)^{\frac{1}{4}} e^{-\alpha \frac{x^2}{2}} \right] \quad (74)$$

$$\frac{d}{dx} e^{-\alpha \frac{x^2}{2}} = -\alpha x e^{-\alpha \frac{x^2}{2}} \quad (75)$$

$$\frac{d^2}{dx^2} e^{-\alpha \frac{x^2}{2}} = (-\alpha + \alpha^2 x^2) e^{-\alpha \frac{x^2}{2}} \quad (76)$$

$$\implies \hat{T}\psi = -\frac{\hbar^2}{2m} (\alpha^2 x^2 - \alpha) \psi \quad (77)$$

For the potential

$$\hat{V}\psi = \left(\frac{1}{2} m\omega^2 x^2 + \lambda_4 x^4 \right) \left[\left(\frac{\alpha}{\pi} \right)^{\frac{1}{4}} e^{-\alpha \frac{x^2}{2}} \right] \quad (78)$$

Now we can find the expectation values

$$\langle \hat{T} \rangle = \left\langle \frac{\hbar^2}{2m} (\alpha - \alpha^2 x^2) \right\rangle = \frac{\hbar^2}{2m} (\alpha - \alpha^2 \langle x^2 \rangle) \quad (79)$$

$$\langle \hat{V} \rangle = \left\langle \left(\frac{1}{2} m \omega^2 x^2 + \lambda_4 x^4 \right) \right\rangle = \frac{1}{2} m \omega^2 \langle x^2 \rangle + \lambda_4 \langle x^4 \rangle \quad (80)$$

We need to find polynomial expectation values

$$\langle x^\tau \rangle = \int_{-\infty}^{\infty} x^\tau |\psi|^2 dx = \left(\frac{\alpha}{\pi} \right)^{\frac{1}{2}} \int_{-\infty}^{\infty} x^\tau e^{-\alpha x^2} dx \quad (81)$$

We can use a handy generalised Gaussian given by

$$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}} \frac{(2n-1)!!}{(2\alpha)^n} \quad (82)$$

Where the double factorial is the same as a single factorial but only keeps integers of the same parity of $2n-1$. We have x^2 and x^4 i.e. $n=1$ and $n=2$.

$$\langle x^2 \rangle = \left(\frac{\alpha}{\pi} \right)^{1/2} \sqrt{\frac{\pi}{\alpha}} \frac{1}{2\alpha} = \frac{1}{2\alpha} \quad (83)$$

$$\langle x^4 \rangle = \left(\frac{\alpha}{\pi} \right)^{1/2} \sqrt{\frac{\pi}{\alpha}} \frac{3}{(4\alpha^2)} = \frac{3}{4\alpha^2} \quad (84)$$

Thus we get the EVs

$$\langle \hat{T} \rangle = \frac{\hbar^2}{2m} \left(\alpha - \alpha^2 \left[\frac{1}{2\alpha} \right] \right) = \frac{\hbar^2 \alpha}{4m} \quad (85)$$

$$\langle \hat{V} \rangle = \left\langle \left(\frac{1}{2} m \omega^2 x^2 + \lambda_4 x^4 \right) \right\rangle = \frac{1}{2} m \omega^2 \left[\frac{1}{2\alpha} \right] + \lambda_4 \left[\frac{3}{4\alpha^2} \right] \quad (86)$$

So we have

$$E_{var} = \frac{\hbar^2 \alpha}{4m} + \frac{1}{4\alpha} m \omega^2 + \frac{3}{4} \frac{\lambda_4}{\alpha^2} \quad (87)$$

(b)

To find the strictest constraint we need the min wrt α .

$$\frac{dE_{var}}{d\alpha} = \frac{\hbar^2}{4m} - \frac{m\omega^2}{4\alpha^2} - \frac{3}{2} \frac{\lambda_4}{\alpha^3} = 0 \quad (88)$$

$$\frac{\hbar^2}{4m} \alpha^3 - \frac{m\omega^2}{4} \alpha - \frac{3}{2} \lambda_4 = 0 \quad (89)$$

Which may be solved explicitly using the cubic formula α^* is the solution.

(c)

With $\lambda_4 = 0$ we retrieve

$$\frac{\hbar^2}{4m} \alpha^3 - \frac{m\omega^2}{4} \alpha = 0 \quad (90)$$

We can take one sol $\alpha^* = 0$ which doesn't satisfy the initial conditions (IC) in the question to get $\frac{\hbar^2}{m} \alpha^2 - m\omega^2 = 0$ (91)

$$\alpha^2 = \frac{m^2 \omega^2}{\hbar^2} \implies \alpha^* = \pm \frac{m\omega}{\hbar} \quad (92)$$

Where we can omit the negative solution as it doesn't satisfy the IC in the question.

$$E_{var}\left(\alpha = \frac{m\omega}{\hbar}\right) = \frac{\hbar^2}{4m}\left(\frac{m\omega}{\hbar}\right) + \frac{1}{4}m\omega^2\left(\frac{\hbar}{m\omega}\right) \quad (93)$$

$$E_{var}(\alpha^*) = \frac{\hbar\omega}{2} \quad (94)$$

Which is the ground state energy for the 1D harmonic oscillator.